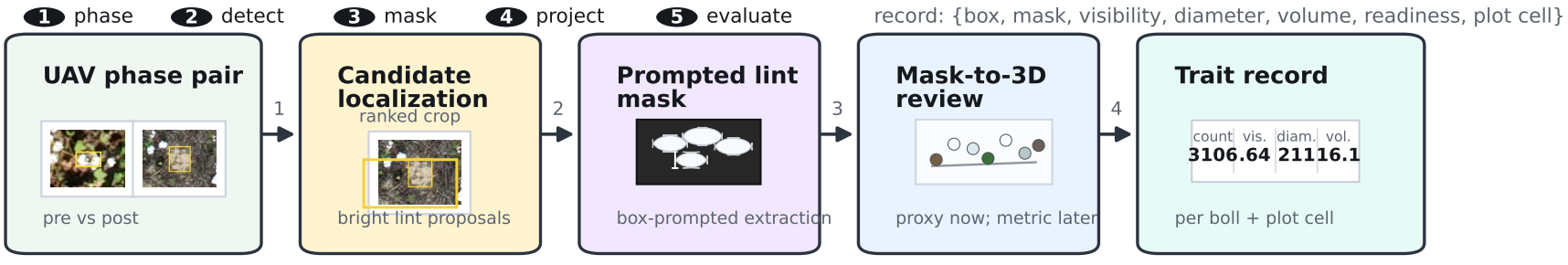
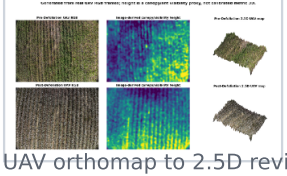
Mask-Guided 3D Cotton Boll Phenotyping

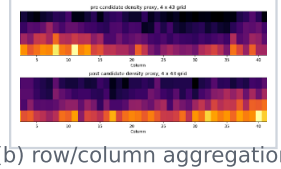
A compact pipeline for pre/post-defoliation UAV imagery, proxy 3D review, and measurement-ready trait evaluation.



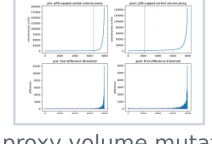
Evidence used in the paper



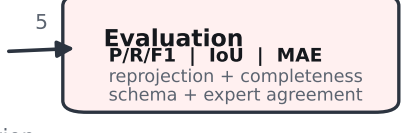
(a) UAV orthomap to 2.5D review



(b) row/column aggregation



(c) proxy volume mutation



Claim boundary Current system produces measurement-ready masks and proxy 2.5D review. Calibrated 3D boll metrology requires GSD/GCP/camera pose, multi-view geometry, RGB-D, or physical measurements.